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System for interrogating serially-connected drug modules in a combinatorial drug delivery device

Abstract

In one aspect, a combinatorial drug delivery device is provided herein including: a plurality of serially connectable modules, each including at least one drug component; and, a master controller having a computing processing unit. Each of the modules includes: a non-transitory memory having stored therein an address and an alphanumeric code representative of the at least one drug component contained in the corresponding drug module; a secondary controller operatively linked to the memory of the corresponding module; and, at least one communication bus connectable in series with the communication buses of the modules serially connected with the corresponding module. The master controller is configured to serially interrogate the secondary controllers over the serially-connected communication buses, using the addresses, to request the codes of the corresponding modules. Advantageously, the subject invention provides for a system to confirm the accuracy of drugs provided with the device and the correct sequencing thereof.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4756706	12/1987	Kerns et al.	N/A	N/A
5807316	12/1997	Teeple, Jr.	N/A	N/A
2004/0171985	12/2003	Schubert et al.	N/A	N/A
2008/0015493	12/2007	Childers et al.	N/A	N/A
2011/0021978	12/2010	Martin et al.	N/A	N/A
2012/0179130	12/2011	Barnes et al.	N/A	N/A
2012/0323208	12/2011	Bochenko et al.	N/A	N/A
2013/0336814	12/2012	Kamen et al.	N/A	N/A
2017/0020784	12/2016	Schweiss et al.	N/A	N/A
2019/0091406	12/2018	Okamura et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0960627	12/1998	EP	N/A
2930710	12/2014	EP	N/A
2018519141	12/2017	JP	N/A
2017537700	12/2019	JP	N/A
1993010851	12/1992	WO	N/A
2012080481	12/2011	WO	N/A
2013102496	12/2012	WO	N/A
2016090315	12/2015	WO	N/A

2019217820	12/2018	WO	N/A
2019217845	12/2018	WO	N/A
2019217864	12/2018	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion of the International Searching Authority from PCT International Application No. PCT/US2020/058507, dated Feb. 22, 2021. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage Application under 35 U.S.C. § 371 of PCT Application No. PCT/US2020/058507, filed Nov. 2, 2020, which claims the priority benefit of U.S. Provisional Application No. 62/932,791, filed Nov. 8, 2019; the contents of which are herein incorporated by reference in their entireties.

BACKGROUND OF THE INVENTION

(1) Combinatorial drug delivery devices and systems are shown and described in: U.S. Provisional Patent Appl. No. 62/670,266, filed May 11, 2018; PCT Appl. No. PCT/US2019/031727, filed May 10, 2019; PCT Appl. No. PCT/US2019/031762, filed May 10, 2019; and, PCT Appl. No. PCT/US2019/031791, filed May 10, 2019. All of the aforementioned patent filings are by the same assignee as herein. As shown in the aforementioned patent filings, drug modules of different liquid drugs may be provided in various combinations to provide different (individualized) drug combinations. The drug modules may be nested, i.e., connected, in series or in parallel, on a tray or other base structure. Alternatively, the drug modules may be serially connected (vertically and/or horizontally) directly to one another. U.S. Provisional Patent Appl. No. 62/670,266, PCT Appl. No. PCT/US2019/031727, PCT Appl. No. PCT/US2019/031762, and, PCT Appl. No. PCT/US2019/031791, are incorporated by reference herein in their respective entireties.

(2) The serially-connected combinatorial system has the advantage in comparison with the nested designs in that it does not require a separate tray component to make the fluid connections and is therefore more efficient in components and, thus, in supply chain.

(3) In the nested system, the tray design can ‘store’ information on the correct configuration of the modules through the inherent design and layout of the tray design. For example, the tray may provide a configuration (e.g., mechanical cooperating features, such as “lock and key” features) that guarantee only the correct drug modules can be inserted into the nests of the tray and that the correct drug modules are arranged in the correct order. This acts as a safety check in preparing the drug modules for use. In contrast, the serially-connected system does not have a tray-type element and, thus, lacks the ability to have a safety check on this basis.

(4) Because tray-based mechanical means of error prevention are not possible in the serially-connected case, it is desirable to implement other means of detecting configuration errors in the serially-connected system and hence prevent the occurrence of medication errors.

SUMMARY OF THE INVENTION

(5) In one aspect, a combinatorial drug delivery device is provided herein including: a plurality of serially connectable modules, each of the modules including at least one drug component; and, a master controller having a computing processing unit. Each of the modules includes: a non-transitory memory having stored therein an address and an alphanumeric code representative of the

at least one drug component contained in the corresponding drug module; a secondary controller operatively linked to the memory of the corresponding module; and, at least one communication bus connectable in series with the communication buses of the modules serially connected with the corresponding module. The master controller is configured to serially interrogate the secondary controllers over the serially-connected communication buses, using the addresses, to request the codes of the corresponding modules. Advantageously, the subject invention provides for a system to confirm the accuracy of drugs provided with the device and the correct sequencing thereof.

(6) In a further aspect, a combinatorial drug delivery device is provided herein including: a plurality of serially connectable modules, each of the modules including at least one drug component; and, a master controller having a computing processing unit. Each of the modules includes: a non-transitory memory having stored therein an alphanumeric code representative of the at least one drug component contained in the corresponding drug module; an actuatable switch having a normally open position, the switch being actuated to a closed position by connecting the corresponding drug module with another of the drug modules; a secondary controller operatively linked to the memory of the corresponding module; and, at least one communication bus connectable in series with the communication buses of the modules serially connected with the corresponding module. With the modules serially connected, each of the secondary controllers determines whether the corresponding switch has been actuated, wherein, upon determining that the corresponding switch has not been actuated, the corresponding secondary controller assigns to the corresponding memory a default address with the default address being transmitted to the master controller. The master controller serially issues commands to the secondary controllers to transmit pulsed signals to adjacent modules, and, wherein, depending on the number of pulsed signals received by the modules, the secondary controllers each assign to the corresponding memory an address with the addresses of the modules being transmitted to the master controller. The master controller is configured to serially interrogate the secondary controller, using the addresses, to request the codes of the corresponding modules.

(7) These and other features of the invention will be better understood through a study of the detailed description and accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIGS. **1-6** depict various features of a device formed in accordance with the subject invention.

DETAILED DESCRIPTION OF THE INVENTION

(2) With reference to FIGS. **1-3**, an arrangement is shown useable to verify the accuracy of a plurality of serially-connected drug modules **12** of a combinatorial drug delivery device **14**. Each of the drug modules **12** includes a drug reservoir **16** for accommodating a liquid drug **18**. The drug reservoirs **16** may be defined by portions of the drug modules **12**, or be defined by components, such as vials, inserted into the drug modules **12**. The combinatorial drug delivery device **14**, including any aspect thereof, may be formed in accordance with any of the embodiments disclosed in any of U.S. Provisional Patent Appl. No. 62/670,266, PCT Appl. No. PCT/US2019/031727, PCT Appl. No. PCT/US2019/031762, and, PCT Appl. No. PCT/US2019/031791. For illustrative purposes, exemplary features of the combinatorial drug delivery device **14** are described herein. As will be recognized by those skilled in the art, the subject invention is useable with any of the combinatorial drug delivery devices, including being useable with any of the elements thereof (e.g., system **10**, drug modules **12**, manner of connecting the drug modules **12**, flow controller **34**, etc.), disclosed in any of the aforementioned patent filings.

(3) As shown in FIG. **1**, the drug modules **12** are serially-connected so as to define a single flow path for the drug delivery device **14** through the series of the drug modules **12**, through which the

liquid drug **18** of each of the drug modules **12** may be drawn. As shown in FIG. **1**, inlet and outlet tubing **20**, **22**, may be provided for each of the drug modules **12** so that the liquid drug **18** may be drawn, in succession, from each of the drug modules **12**. As shown in FIG. **3**, the inlet and outlet tubing **20**, **22** may be formed continuously between the drug reservoirs **16** so that lengths of tubing are provided serving both as an outlet of one of the drug reservoirs **16** and an inlet for the next drug reservoir **16**. FIG. **1** shows six of the drug modules **12** (**12A-12F**). As will be appreciated by those skilled in the art, any quantity of the drug modules **12** may be utilized. A vent **13** may be provided at a terminus of the flow path (in the ultimate drug module).

(4) It is noted that one or more by-pass drug modules **12BY** may be needed in a series, to accommodate a place in the series, but to not contain any liquid drug. As shown in FIG. **2**, the by-pass drug module **12BY** may have by-pass tubing **24** which extends from the inlet to the outlet thereof to allow for flow therethrough without a drug reservoir. Alternatively, as shown in FIG. **3**, the by-pass tubing **24** may be provided in lieu of one of the drug modules **12** to connect two components of the drug delivery device **14**, such as two of the drug modules **12** or one of the drug modules **12** and the controller housing described below.

(5) The liquid drugs **18** contained in the drug modules **12** may vary in type and concentration. The liquid drug **18** in some of the modules **12** may be a diluent with no pharmaceutically or biologically active agents. The drug modules **12** may contain one or more solid components which can be reconstituted with flow of a diluent therein to form a liquid drug. The ability of the serially-connected drug modules **12** to contain various drug types and concentrations allows for the drug delivery device **14** to be a combinatorial drug delivery device **14**, providing for the mixing of various liquid drugs. The liquid drugs **18** intended for a particular combination for a patient is prescribed by a physician. The subject invention provides for the confirmation of accuracy of the inclusion of the particular drug modules **12** in the drug delivery device **14**, as well as, the sequence of the drug modules **12**. The sequence of the drug modules **14** may be significant, possibly having an impact on the efficacy of the ultimate resulting combination.

(6) The drug delivery device **14** preferably includes a controller housing **26** to which the serially-connected drug modules **12** are connected. The outlet tubing **22** of the first drug module **12A** (being the closest to the controller housing **26**) is in communication with an inlet **28** formed in the controller housing **26** into which the liquid drug **18** may flow from the drug modules **12**. Delivery tubing **30** extends from the inlet **28** to convey the liquid drug **18** through the controller housing **26** to an outlet **32**. Tubing or conveyances may be secured to the outlet **32** to direct the liquid drug **18** to a storage device (e.g., an IV bag, injector) or to a drug delivery device connected to a patient (e.g., a butterfly needle).

(7) A flow controller **34** is provided in the controller housing **26** which selectively regulates flow through the delivery tubing **30**. In one embodiment, the flow controller **34** may include an actuatable source of negative pressure **36**, such as a pump, provided in the controller housing **26** to draw the liquid drug **18** through the inlet **28** and discharge the liquid drug **18** through the outlet **32**, via the delivery tubing **30** (which may be discontinuous). In a quiescent state, the source of negative pressure **36** generates no negative pressure, thus, not drawing the liquid drug **18**. In a further embodiment, the flow controller **34** may include one or more adjustable valves **38** provided in the controller housing **26** configured to selectively regulate flow through the delivery tubing **30**, particularly being configured to be selectively adjusted between open and closed states, such as a ball valve. With the use of the valves **38**, a source of negative pressure external to the controller housing **26** may be utilized which is configured to apply negative pressure to the outlet **32** to draw the liquid drug **18** therefrom.

(8) A control unit **40** may be provided in the controller housing **26** which includes a computing processing unit (CPU) **42**. It is preferred that the flow controller **34** be electrically powered to be controlled by the CPU **42**. For example, an electrical motor or actuator may be provided having a switch configured to be controlled by the CPU **42**. Actuation of the motor can cause the source of

negative pressure **30** to be activated (e.g., the pump to be turned on), whereas, actuation of the actuator can cause adjustment of the valve(s) **38** to an open state (e.g., rotation of the valve stem to an open state). The switch may be adjusted to an off position by the CPU **42** to turn off the motor, or close the valve(s).

(9) It is envisioned that the drug modules **12** will be serially-connected, when ready for use. Thus, assembly of the drug modules **12** is required by a user, or on behalf of a user. As a fail-safe mechanism, to ensure that the drug modules **12** are properly included in the drug delivery device **14** and in the correct sequence, each of the drug modules **12** may have an alphanumeric code stored therein representative of the drugs **18** contained in the drug modules **12**. The alphanumeric codes may designate a drug type and, possibly, a drug's concentration or strength. The liquid drug **18** may be loaded into the drug modules **12** in a manufacturing facility or in a pharmacy with the alphanumeric codes being stored on the modules **12** at the same time. Care is needed to store the correct alphanumeric codes in the drug modules **12**.

(10) The specific liquid drugs **18** (type, concentration) will be specified by prescription. The drug modules **12** will be prepared to accommodate the specified liquid drugs **18**—the number of the drug modules **12** to be utilized being at least equal to the number of drug components specified by the prescription. The drug modules **12**, along with the controller housing **26**, may be delivered to the user or a location associated with the user as a kit, for assembly. Instructions will be provided with regards to the assembly of the drug modules **12**, including the sequence of the drug modules **12**, e.g., first position (closest to the controller housing **26**), second position, and so forth.

(11) With reference to FIG. **4**, a secondary controller **1**, which may be a microprocessor or a programmable logic device, is provided for each of the modules **12**. The secondary controller **1** may run embedded firmware to implement the logic described below. The secondary controller **1** has access to a compact serial communication bus **2**, such as I^{sup}.2C (inter-integrated circuit), as known in embedded electronics. This interface requires two lines: SCL (serial clock) and SDA (serial data). With the modules **12** being serially connected, the communication buses **2** of the modules **12** are serially connected to allow for communication signals to pass between the modules **12**. In addition, the communication bus **2** of the first drug module **12A** is communicatively coupled with the control unit **40**, which acts as a master controller. A non-transitory memory **4**, such as an address register or other persistent memory, is operatively linked with the secondary controller **1** of each of the modules **12**. The memory **4** may be programmed with a unique address representing a position of the corresponding module **12** (e.g., first position for the module **12A**, second position for the module **12B**, and so forth). The memory **4** may be used to store the alphanumeric code of the corresponding drugs **18**. The unique address may be also stored in the memory **4** at the same time as the alphanumeric code. With the modules **12** being serially connected, as shown in FIG. **5**, the control unit **40** may serially interrogate the secondary controllers **1** of the modules **12** over the serially connected communication buses **2**, using the unique addresses, to request the stored alphanumeric codes of each of the modules **12**. The secondary controllers **1** may transmit the alphanumeric codes, via the serially connected communication buses **2**, to the control unit **40**. In this way, small volumes of data on the order of a few bytes can be quickly and compactly communicated to the control unit **40**. The alphanumeric codes may be placed in sequence to generate an activation code.

(12) The activation code may be used for comparison against an authentication code to determine its accuracy. In one embodiment, the authentication code may be stored in a non-transitory memory **41** associated with the CPU **42** in the controller housing **26**. Alternatively, the authentication code may be transmitted to the CPU **42** (e.g., via a receiver on the controller housing **26**) with the CPU **42** running a comparison to determine a match. With a match between the activation code and the authentication code, the CPU **42** may actuate the flow controller **34** to enable the delivery of the liquid drug **18**.

(13) The flow controller **34** may be provided to have a storage (i.e., non-use) state, e.g., where one

or more of the adjustable valves **38** are in closed positions to not permit flow through the delivery tubing **30** to the outlet **32**. In addition, or alternatively, in the storage state, the source of negative pressure **36** is in a quiescent state. With a match of the activation code and the authentication code, as described above, the CPU **42** may actuate the flow controller **34**, thus causing the flow controller **34** to enter a use state. With the flow controller **34** in a use state, delivery of the liquid drug **18** from the drug delivery device **14** may be achieved. In particular, the one or more adjustable valves **38** may be adjusted to open positions to permit flow through the delivery tubing **30** to the outlet **32**. In addition, the source of negative pressure **36** may be actuated, or, alternatively, may be placed into an active state, awaiting actuation (e.g., by a switch on the controller housing **26**).

(14) In alternate embodiment, the modules **12** may be configured to self-assign a unique address based on positioning within the device **14**. In this embodiment, the modules **12** are not initially provided with an address in the memory **4**. To implement this scheme, each of the modules **12** is provided with an actuatable switch **3** (e.g., a mechanical switch) which is preferably normally open. The switches **3** are provided on the modules **12** to be only actuated (i.e., actuated to a closed position) when a module has another module attached to it downstream. In addition, for each of the modules **12**, an input line **5a** is provided, to link with the downstream module **12**, that the secondary controller **1** can read, and an output line **5b** is provided to link with the upstream module **12**, that the secondary controller **1** can control.

(15) In this embodiment, with the modules **12** being serially connected, the most downstream module **12** (**12F**) is the only module that does not have its mechanical switch actuated. The secondary controller **1** on this module **12** therefore concludes that its address should be 1 and assigns itself that address by writing to its memory **4**. This address is transmitted to the control unit **40** using the serially connected communication buses **2**. The control unit **40**, knowing that the module **12** with Address **1** will have automatically assigned itself that address, can then freely communicate with that module over the serially connected communication buses **2**.

(16) To establish the addresses of the other modules **12**, the control unit **40** sends a command to Address **1**, i.e., the module **12F**, telling it to assert its upstream line **5b** two times. The module **12F** with Address **1** performs this pulsing sequence, which the next upstream module **12E** sees by virtue of its input line **5a**. Because it sees two pulses, the module **12E** can conclude that it must be Address **2**. The module **12E** then writes that address to its memory **4**. Subsequently, the control unit **40** knows that Address **2** has been assigned and can freely communicate with that module. So, the control unit **40** issues a command to Address **2**, i.e., the module **12E**, to pulse its output line **5b** three times. The next module **12D** up sees this pulsing sequence, counting the pulses and concluding that it must be Address **3**. This process continues until all addresses have been assigned. Then, the control unit **40** can query each address for its alphanumeric code as explained above. This address assignment protocol is explicated sequentially in FIG. **6**.

(17) In one embodiment, any of the combinatorial drug delivery devices disclosed herein is able to deliver two or more drugs for the benefit of the patient suffering from any of a wide range of diseases or conditions, e.g., cancer, autoimmune disorder, inflammatory disorder, cardiovascular disease or fibrotic disorder. In one embodiment, one or more of drug modules **12** may contain a single drug. In one embodiment, one or more of drug module **12** may contain two or more co-formulated drugs. In one embodiment, one or more of drug module **12** may contain a drug in solid form (such as a tablet, capsule, powder, lyophilized, spray dried), which can be reconstituted with flow of a diluent therein to form a liquid drug.

(18) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an immune checkpoint inhibitor. In certain embodiments, the immune checkpoint inhibitor is Programmed Death-1 (“PD-1”) pathway inhibitor, a cytotoxic T-lymphocyte-associated antigen 4 (“CTLA-4”) antagonist, a Lymphocyte Activation Gene-3 (“LAG3”) antagonist, a CD80 antagonist, a CD86 antagonist, a T cell immunoglobulin and mucin domain (“Tim-3”) antagonist, a T cell immunoreceptor with Ig and ITIM domains (“TIGIT”)

antagonist, a CD20 antagonist, a CD96 antagonist, a indoleamine 2,3-dioxygenase (“IDO1”) antagonist, a stimulator of interferon genes (“STING”) antagonist, a GARP antagonist, a CD40 antagonist, Adenosine A2A receptor (“A2aR”) antagonist, a CEACAM1 (CD66a) antagonist, a CEA antagonist, a CD47 antagonist, a Receptor Related Immunoglobulin Domain Containing Protein (“PVRIG”) antagonist, a tryptophan 2,3-dioxygenase (“TDO”) antagonist, a V-domain Ig suppressor of T cell activation (“VISTA”) antagonist, or a Killer-cell Immunoglobulin-like Receptor (“KIR”) antagonist.

(19) In one embodiment, the PD-1 pathway inhibitor is an anti-PD-1 antibody or antigen binding fragment thereof. In certain embodiments, the anti-PD-1 antibody is pembrolizumab (KEYTRUDA; MK-3475), pidilizumab (CT-011), nivolumab (OPDIVO; BMS-936558), PDR001, MEDI0680 (AMP-514), TSR-042, REGN2810, JS001, AMP-224 (GSK-2661380), PF-06801591, BGB-A317, BI 754091, or SHR-1210.

(20) In one embodiment, the PD-1 pathway inhibitor is an anti-PD-L1 antibody or antigen binding fragment thereof. In certain embodiments, the anti-PD-L1 antibody is atezolizumab (TECENTRIQ; RG7446; MPDL3280A; RO5541267), durvalumab (MEDI4736), BMS-936559, avelumab (bavencio), LY3300054, CX-072 (Proclaim-CX-072), FAZ053, KN035, or MDX-1105.

(21) In one embodiment, the PD-1 pathway inhibitor is a small molecule drug. In certain embodiments, the PD-1 pathway inhibitor is CA-170. In another embodiment, the PD-1 pathway inhibitor is a cell based therapy. In one embodiment, the cell based therapy is a MiHA-loaded PD-L1/L2-silenced dendritic cell vaccine. In other embodiments, the cell based therapy is an anti-programmed cell death protein 1 antibody expressing pluripotent killer T lymphocyte, an autologous PD-1-targeted chimeric switch receptor-modified T lymphocyte, or a PD-1 knockout autologous T lymphocyte.

(22) In one embodiment, the PD-1 pathway inhibitor is an anti-PD-L2 antibody or antigen binding fragment thereof. In another embodiment, the anti-PD-L2 antibody is rHIgM12B7.

(23) In one embodiment, the PD-1 pathway inhibitor is a soluble PD-1 polypeptide. In certain embodiments, the soluble PD-1 polypeptide is a fusion polypeptide. In some embodiments, the soluble PD-1 polypeptide comprises a ligand binding fragment of the PD-1 extracellular domain. In other embodiments, the soluble PD-1 polypeptide comprises a ligand binding fragment of the PD-1 extracellular domain. In another embodiment, the soluble PD-1 polypeptide further comprises an Fc domain.

(24) In one embodiment, the immune checkpoint inhibitor is a CTLA-4 antagonist. In certain embodiments, the CTLA-4 antagonist is an anti-CTLA-4 antibody or antigen binding fragment thereof. In some embodiments, the anti-CTLA-4 antibody is ipilimumab (YERVOY), tremelimumab (ticilimumab; CP-675,206), AGEN-1884, or ATOR-1015. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a CTLA-4 antagonist, e.g., ipilimumab (YERVOY), and a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA).

(25) In one embodiment, the immune checkpoint inhibitor is an antagonist of LAG3. In certain embodiments, the LAG3 antagonist is an anti-LAG3 antibody or antigen binding fragment thereof. In certain embodiments, the anti-LAG3 antibody is relatlimab (BMS-986016), MK-4280 (28G-10), REGN3767, GSK2831781, IMP731 (H5L7BW), BAP050, IMP-701 (LAG-5250), IMP321, TSR-033, LAG525, BI 754111, or FS-118. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a LAG3 antagonist, e.g., relatlimab or MK-4280, and a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA). In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a LAG3 antagonist, e.g., relatlimab or MK-4280, and a CTLA-4 antagonist, e.g., ipilimumab (YERVOY). In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a LAG3 antagonist, e.g., relatlimab or MK-4280, a CTLA-4 antagonist, e.g., ipilimumab (YERVOY), and a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab

(KEYTRUDA).

(26) In one embodiment, the immune checkpoint inhibitor is a KIR antagonist. In certain embodiments, the KIR antagonist is an anti-KIR antibody or antigen binding fragment thereof. In some embodiments, the anti-KIR antibody is lirilumab (1-7F9, BMS-986015, IPH 2101) or IPH4102.

(27) In one embodiment, the immune checkpoint inhibitor is TIGIT antagonist. In one embodiment, the TIGIT antagonist is an anti-TIGIT antibody or antigen binding fragment thereof. In certain embodiments, the anti-TIGIT antibody is BMS-986207, AB 154, COM902 (CGEN-15137), or OMP-313M32.

(28) In one embodiment, the immune checkpoint inhibitor is Tim-3 antagonist. In certain embodiments, the Tim-3 antagonist is an anti-Tim-3 antibody or antigen binding fragment thereof. In some embodiments, the anti-Tim-3 antibody is TSR-022 or LY3321367.

(29) In one embodiment, the immune checkpoint inhibitor is an IDO1 antagonist. In another embodiment, the IDO1 antagonist is indoximod (NLG8189; 1-methyl-D-TRP), epacadostat (INCB-024360, INCB-24360), KHK2455, PF-06840003, navoximod (RG6078, GDC-0919, NLG919), BMS-986205 (F001287), or pyrrolidine-2,5-dione derivatives.

(30) In one embodiment, the immune checkpoint inhibitor is a STING antagonist. In certain embodiments, the STING antagonist is 2' or 3'-mono-fluoro substituted cyclic-di-nucleotides; 2'3'-di-fluoro substituted mixed linkage 2',5'-3',5' cyclic-di-nucleotides; 2'-fluoro substituted, bis-3',5' cyclic-di-nucleotides; 2',2''-diF-Rp,Rp,bis-3',5' cyclic-di-nucleotides; or fluorinated cyclic-di-nucleotides.

(31) In one embodiment, the immune checkpoint inhibitor is CD20 antagonist. In some embodiments, the CD20 antagonist is an anti-CD20 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD20 antibody is rituximab (RITUXAN; IDEC-102; IDEC-C2B8), ABP 798, ofatumumab, or obinutuzumab.

(32) In one embodiment, the immune checkpoint inhibitor is CD80 antagonist. In certain embodiments, the CD80 antagonist is an anti-CD80 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD80 antibody is galiximab or AV 1142742.

(33) In one embodiment, the immune checkpoint inhibitor is a GARP antagonist. In some embodiments, the GARP antagonist is an anti-GARP antibody or antigen binding fragment thereof. In certain embodiments, the anti-GARP antibody is ARGX-115.

(34) In one embodiment, the immune checkpoint inhibitor is a CD40 antagonist. In certain embodiments, the CD40 antagonist is an anti-CD40 antibody or antigen binding fragment thereof. In some embodiments, the anti-CD40 antibody is BMS3h-56, lucatumumab (HCD122 and CHIR-12.12), CHIR-5.9, or dacetuzumab (huS2C6, PRO 64553, RG 3636, SGN 14, SGN-40). In another embodiment, the CD40 antagonist is a soluble CD40 ligand (CD40-L). In one embodiment, the soluble CD40 ligand is a fusion polypeptide. In one embodiment, the soluble CD40 ligand is a CD40-L/FC2 or a monomeric CD40-L.

(35) In one embodiment, the immune checkpoint inhibitor is an A2aR antagonist. In some embodiments, the A2aR antagonist is a small molecule. In certain embodiments, the A2aR antagonist is CPI-444, PBF-509, istradefylline (KW-6002), preladenant (SCH420814), tozadenant (SYN115), vipadenant (BIIB014), HTL-1071, ST1535, SCH412348, SCH442416, SCH58261, ZM241385, or AZD4635.

(36) In one embodiment, the immune checkpoint inhibitor is a CEACAM1 antagonist. In some embodiments, the CEACAM1 antagonist is an anti-CEACAM1 antibody or antigen binding fragment thereof. In one embodiment, the anti-CEACAM1 antibody is CM-24 (MK-6018).

(37) In one embodiment, the immune checkpoint inhibitor is a CEA antagonist. In one embodiment, the CEA antagonist is an anti-CEA antibody or antigen binding fragment thereof. In certain embodiments, the anti-CEA antibody is cergutuzumab amunaleukin (RG7813, RO-6895882) or RG7802 (RO6958688).

(38) In one embodiment, the immune checkpoint inhibitor is a CD47 antagonist. In some embodiments, the CD47 antagonist is an anti-CD47 antibody or antigen binding fragment thereof. In certain embodiments, the anti-CD47 antibody is HuF9-G4, CC-90002, TTI-621, ALX148, NI-1701, NI-1801, SRF231, or Effi-DEM.

(39) In one embodiment, the immune checkpoint inhibitor is a PVRIG antagonist. In certain embodiments, the PVRIG antagonist is an anti-PVRIG antibody or antigen binding fragment thereof. In one embodiment, the anti-PVRIG antibody is COM701 (CGEN-15029).

(40) In one embodiment, the immune checkpoint inhibitor is a TDO antagonist. In one embodiment, the TDO antagonist is a 4-(indol-3-yl)-pyrazole derivative, a 3-indol substituted derivative, or a 3-(indol-3-yl)-pyridine derivative. In another embodiment, the immune checkpoint inhibitor is a dual IDO and TDO antagonist. In one embodiment, the dual IDO and TDO antagonist is a small molecule.

(41) In one embodiment, the immune checkpoint inhibitor is a VISTA antagonist. In some embodiments, the VISTA antagonist is CA-170 or JNJ-61610588.

(42) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an immune checkpoint enhancer or stimulator.

(43) In one embodiment, the immune checkpoint enhancer or stimulator is a CD28 agonist, a 4-1BB agonist, an OX40 agonist, a CD27 agonist, a CD80 agonist, a CD86 agonist, a CD40 agonist, an ICOS agonist, a CD70 agonist, or a GITR agonist.

(44) In one embodiment, the immune checkpoint enhancer or stimulator is an OX40 agonist. In certain embodiments, the OX40 agonist is an anti-OX40 antibody or antigen binding fragment thereof. In some embodiments, the anti-OX40 antibody is tavolixizumab (MEDI-0562), pogalizumab (MOXR0916, RG7888), GSK3174998, ATOR-1015, MEDI-6383, MEDI-6469, BMS 986178, PF-04518600, or RG7888 (MOXR0916). In another embodiment, the OX40 agonist is a cell based therapy. In certain embodiments, the OX40 agonist is a GINAKIT cell (iC9-GD2-CD28-OX40-expressing T lymphocytes).

(45) In one embodiment, the immune checkpoint enhancer or stimulator is a CD40 agonist. In some embodiments, the CD40 agonist is an anti-CD40 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD40 antibody is ADC-1013 (JNJ-64457107), RG7876 (RO-7009789), HuCD40-M2, APX005M (EPI-0050), or Chi Lob 7/4. In another embodiment, the CD40 agonist is a soluble CD40 ligand (CD40-L). In one embodiment, the soluble CD40 ligand is a fusion polypeptide. In certain embodiments, the soluble CD40 ligand is a trimeric CD40-L (AVREND®).

(46) In one embodiment, the immune checkpoint enhancer or stimulator is a GITR agonist. In certain embodiments, the GITR agonist is an anti-GITR antibody or antigen binding fragment thereof. In one embodiment, the anti-GITR antibody is BMS-986156, TRX518, GWN323, INCAGN01876, or MEDI1873. In one embodiment, the GITR agonist is a soluble GITR ligand (GITRL). In some embodiments, the soluble GITR ligand is a fusion polypeptide. In another embodiment, the GITR agonist is a cell based therapy. In one embodiment, the cell based therapy is an anti-CTLA4 mAb RNA/GITRL RNA-transfected autologous dendritic cell vaccine or a GITRL RNA-transfected autologous dendritic cell vaccine.

(47) In one embodiment, the immune checkpoint enhancer or stimulator is a 4-1BB agonist. In some embodiments, the 4-1BB agonist is an anti-4-1BB antibody or antigen binding fragment thereof. In one embodiment, the anti-4-1BB antibody is urelumab or PF-05082566.

(48) In one embodiment, the immune checkpoint enhancer or stimulator is a CD80 agonist or a CD86 agonist. In some embodiments, the CD80 agonist or the CD86 agonist is a soluble CD80 or CD86 ligand (CTLA-4). In certain embodiments, the soluble CD80 or CD86 ligand is a fusion polypeptide. In one embodiment, the CD80 or CD86 ligand is CTLA4-Ig (CTLA4-IgG4m, RG2077, or RG1046) or abatacept (ORENCIA, BMS-188667). In other embodiments, the CD80 agonist or the CD86 agonist is a cell based therapy. In one embodiment, the cell based therapy is MGN1601 (an allogeneic renal cell carcinoma vaccine).

(49) In one embodiment, the immune checkpoint enhancer or stimulator is a CD28 agonist. In some embodiments, the CD28 agonist is an anti-CD28 antibody or antigen binding fragment thereof. In certain embodiments, the anti-CD28 antibody is TGN1412.

(50) In one embodiment, the CD28 agonist is a cell based therapy. In certain embodiments, the cell based therapy is JCAR015 (anti-CD19-CD28-zeta modified CAR CD3⁺ T lymphocyte); CD28CAR/CD137CAR-expressing T lymphocyte; allogeneic CD4⁺ memory Th1-like T cell s/microparticle-bound anti-CD3/anti-CD28; anti-CD19/CD28/CD3zeta CAR gammaretroviral vector-transduced autologous T lymphocytes KTE-C19; anti-CEA IgCD28TCR-transduced autologous T lymphocytes; anti-EGFRvIII CAR-transduced allogeneic T lymphocytes; autologous CD123CAR-CD28-CD3zeta-EGFRt-expressing T lymphocytes; autologous CD171-specific CAR-CD28 zeta-4-1-BB-EGFRt-expressing T lymphocytes; autologous CD19CAR-CD28-CD3zeta-EGFRt-expressing Tcm-enriched T cells; autologous PD-1-targeted chimeric switch receptor-modified T lymphocytes (chimera with CD28); CD19CAR-CD28-CD3zeta-EGFRt-expressing Tcm-enriched T lymphocytes; CD19CAR-CD28-CD3zeta-EGFRt-expressing Tn/mem-enriched T lymphocytes; CD19CAR-CD28zeta-4-1BB-expressing allogeneic T lymphocytes; CD19CAR-CD3zeta-4-1BB-CD28-expressing autologous T lymphocytes; CD28CAR/CD137CAR-expressing T lymphocytes; CD3/CD28 costimulated vaccine-primed autologous T lymphocytes; or iC9-GD2-CD28-OX40-expressing T lymphocytes.

(51) In one embodiment, the immune checkpoint enhancer or stimulator is a CD27 agonist. In certain embodiments, the CD27 agonist is an anti-CD27 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD27 antibody is varlilumab (CDX-1127).

(52) In one embodiment, the immune checkpoint enhancer or stimulator is a CD70 agonist. In some embodiments, the CD70 agonist is an anti-CD70 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD70 antibody is ARGX-110.

(53) In one embodiment, the immune checkpoint enhancer or stimulator is an ICOS agonist. In certain embodiments, the ICOS agonist is an anti-ICOS antibody or antigen binding fragment thereof. In some embodiments, the anti-ICOS antibody is BMS986226, MEDI-570, GSK3359609, or JTX-2011. In other embodiments, the ICOS agonist is a soluble ICOS ligand. In some embodiments, the soluble ICOS ligand is a fusion polypeptide. In one embodiment, the soluble ICOS ligand is AMG 750.

(54) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an anti-CD73 antibody or antigen binding fragment thereof. In certain embodiments, the anti-CD73 antibody is MEDI9447.

(55) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is a TLR9 agonist. In one embodiment, the TLR9 agonist is agatolimod sodium.

(56) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is a cytokine. In certain embodiments, the cytokine is a chemokine, an interferon, an interleukin, lymphokine, or a member of the tumor necrosis factor family. In some embodiments, the cytokine is IL-2, IL-15, or interferon-gamma.

(57) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is a TGF- β antagonist. In some embodiments, the TGF- β antagonist is fresolimumab (GC-1008); NIS793; IMC-TR1 (LY3022859); ISTH0036; trabedersen (AP 12009); recombinant transforming growth factor-beta-2; autologous HPV-16/18 E6/E7-specific TGF-beta-resistant T lymphocytes; or TGF-beta-resistant LMP-specific cytotoxic T-lymphocytes.

(58) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an iNOS antagonist. In some embodiments, the iNOS antagonist is N-Acetylcysteine (NAC), aminoguanidine, L-nitroarginine methyl ester, or S,S-1,4-phenylenebis(1,2-ethanediy)bis-isothiourea).

(59) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery

devices disclosed herein is a SHP-1 antagonist.

(60) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is a colony stimulating factor 1 receptor (“CSF1R”) antagonist. In certain embodiments, the CSF1R antagonist is an anti-CSF1R antibody or antigen binding fragment thereof. In some embodiments, the anti-CSF1R antibody is emactuzumab.

(61) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an agonist of a TNF family member. In some embodiments, the agonist of the TNF family member is ATOR 1016, ABBV-621, or Adalimumab.

(62) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is an Interleukin-2 (IL-2), such as aldesleukin. Preferably, the IL-2 or conjugated IL-2 (e.g., pegylated) has been modified to selectively activate T-effector cells over T-regulatory cells (“T-eff IL-2”), such as bempegaldesleukin. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, and a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA). In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, and a LAG3 antagonist, e.g., relatlimab or MK-4280. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, and a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA), and a LAG3 antagonist, e.g., relatlimab or MK-4280. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells and a CTLA-4 antagonist, e.g., ipilimumab (YERVOY). In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA), and a CTLA-4 antagonist, e.g., ipilimumab (YERVOY). In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, a CTLA-4 antagonist, e.g., ipilimumab (YERVOY), and a LAG3 antagonist, e.g., relatlimab or MK-4280. In one embodiment, any of the combinatorial drug delivery devices disclosed herein includes a modified IL-2, such as bempegaldesleukin, which selectively activates T-effector cells over T-regulatory cells, a PD-1 pathway inhibitor, e.g., nivolumab (OPDIVO) or pembrolizumab (KEYTRUDA), a CTLA-4 antagonist, e.g., ipilimumab (YERVOY), and a LAG3 antagonist, e.g., relatlimab or MK-4280.

(63) In one embodiment, one or more of the drugs of any of the combinatorial drug delivery devices disclosed herein is a CD160 (NK1) agonist. In certain embodiments, the CD160 (NK1) agonist is an anti-CD160 antibody or antigen binding fragment thereof. In one embodiment, the anti-CD160 antibody is BY55.

(64) In one embodiment, the one or more of drug module **12** may contain a soluble CTLA-4 polypeptide, which can be useful for treating, for instance, T-cell mediated autoimmune disorders, such as rheumatoid arthritis, juvenile idiopathic arthritis, psoriatic arthritis, graft-versus-host disease, and transplant rejection. In one embodiment, the soluble CTLA-4 polypeptide is abatacept (ORENCIA), belatacept (NULOJIX), RG2077, or RG-1046. In certain embodiments, one or more drug modules **12** of a combinatorial drug delivery device as described herein include a soluble CTLA-4 polypeptide, e.g., abatacept (ORENCIA) and a Bruton's tyrosine kinase inhibitor, e.g., branebrutinib. In certain embodiments, one or more drug modules **12** of a combinatorial drug delivery device as described herein include a soluble CTLA-4 polypeptide, e.g., abatacept (ORENCIA) and a tyrosine kinase-2 inhibitor, e.g., BMS-986165. In certain embodiments, one or

more drug modules 12 of a combinatorial drug delivery device as described herein include a soluble CTLA-4 polypeptide, e.g., abatacept (ORENCIA) and an Interleukin-2 (IL-2) or “T-reg IL-2”, which selectively activates T-regulatory cells as opposed to T-effector cells, e.g., BMS-986326 and NKTR-358.

Claims

1. A combinatorial drug delivery device comprising: a plurality of serially connectable modules, each of the modules including at least one drug component, wherein the modules are serially connectable to define a single flow path for the drug delivery device; and, a master controller having a computing processing unit, wherein, each of the modules includes: a non-transitory memory having stored therein an address and an alphanumeric code representative of the at least one drug component contained in the corresponding drug module; a secondary controller operatively linked to the memory of the corresponding module; at least one communication bus connectable in series with the communication buses of the modules serially connected with the corresponding module, wherein, with the modules being serially connected, the serially-connected communication buses of the modules allow for communication signals to directly pass between the modules, wherein, the master controller is configured to serially interrogate the secondary controllers over the serially-connected communication buses, using the addresses, to request the codes of the corresponding modules.
2. A combinatorial drug delivery device as in claim 1, wherein the master controller is configured to place the received codes of the corresponding modules in sequence to generate an activation code.
3. A combinatorial drug delivery device as in claim 2, wherein the master controller is configured to compare the generated activation code with an authentication code.
4. A combinatorial drug delivery device as in claim 3, further comprising an actuatable flow controller to selectively regulate flow of the at least one drug component, wherein, the master controller is configured to actuate the flow controller with a match of the generated activation code and the authentication code.
5. A combinatorial drug delivery device as in claim 4, wherein, with actuation, the flow controller causes at least one of: adjustment of one or more adjustable values; actuation of a source of negative pressure; and, placing a source of negative pressure into an active state awaiting actuation.
6. A combinatorial drug delivery device comprising: a plurality of serially connectable modules, each of the modules including at least one drug component; and, a master controller having a computing processing unit, wherein, each of the modules includes: a non-transitory memory having stored therein an alphanumeric code representative of the at least one drug component contained in the corresponding drug module; an actuatable switch having a normally open position, the switch being actuated to a closed position by connecting the corresponding drug module with another of the drug modules; a secondary controller operatively linked to the memory of the corresponding module; at least one communication bus connectable in series with the communication buses of the modules serially connected with the corresponding module, wherein, with the modules serially connected, each of the secondary controllers determines whether the corresponding switch has been actuated, wherein, upon determining that the corresponding switch has not been actuated, the corresponding secondary controller assigns to the corresponding memory a default address with the default address being transmitted to the master controller, wherein, the master controller serially issues commands to the secondary controllers to transmit pulsed signals to adjacent modules, and, wherein, depending on the number of pulsed signals received by the modules, the secondary controllers each assign to the corresponding memory an address with the addresses of the modules being transmitted to the master controller, wherein, the master controller is configured to serially interrogate the secondary controllers, using the addresses, to request the codes of the corresponding

modules.

7. A combinatorial drug delivery device as in claim 6, wherein the master controller is configured to place the received codes of the corresponding modules in sequence to generate an activation code.

8. A combinatorial drug delivery device as in claim 7, wherein the master controller is configured to compare the generated activation code with an authentication code.

9. A combinatorial drug delivery device as in claim 8, further comprising an actuatable flow controller to selectively regulate flow of the at least one drug component, wherein, the master controller is configured to actuate the flow controller with a match of the generated activation code and the authentication code.

10. A combinatorial drug delivery device as in claim 9, wherein, with actuation, the flow controller causes at least one of: adjustment of one or more adjustable values; actuation of a source of negative pressure; and, placing a source of negative pressure into an active state awaiting actuation.
